

Supervised Learning

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Classification of ML algorithms

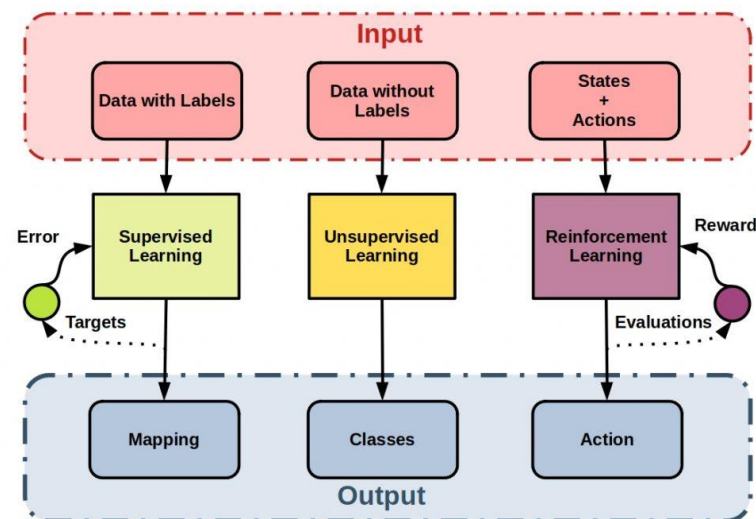


Image source: <https://starship-knowledge.com/supervised-vs-unsupervised-vs-reinforcement>

- Supervised learning: Using labeled data to train models.
- Unsupervised learning: No labeled data is needed.
- Reinforcement learning: Intelligent agents perform actions so as to maximize cumulative reward.

Classification of ML algorithms

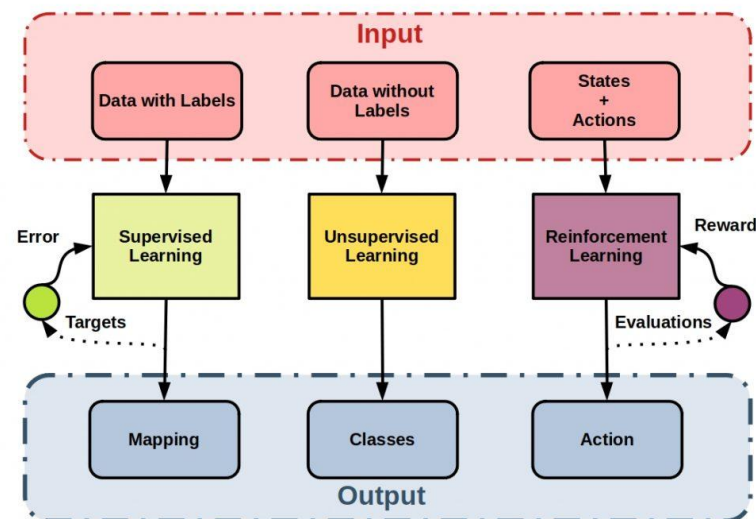


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- **Supervised learning:** Using labeled data to train models.
- **Unsupervised learning:** No labeled data is needed.
- **Reinforcement learning:** Intelligent agents perform actions so as to maximize cumulative reward.

Unsupervised learning

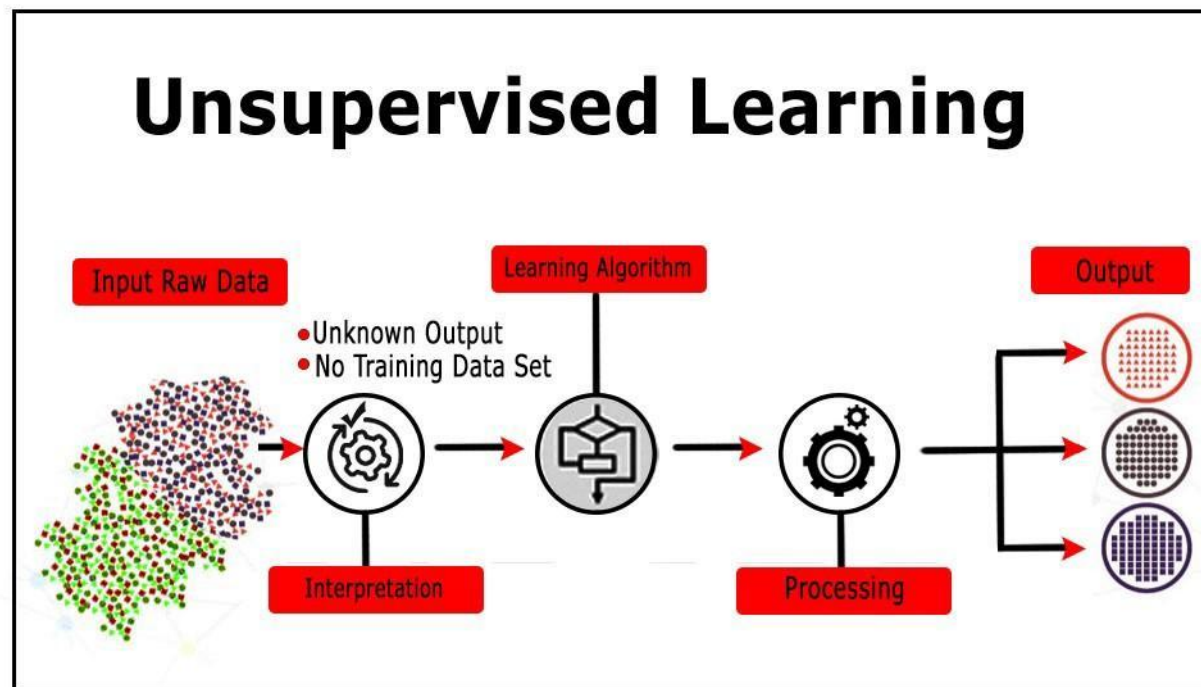


Image source: <https://medium.datadriveninvestor.com/supervised-and-unsupervised-learning-7281050992a0>

- Automatic extraction of patterns from data by analyzing the similarity among instances.
- Application: Clustering.

Unsupervised learning

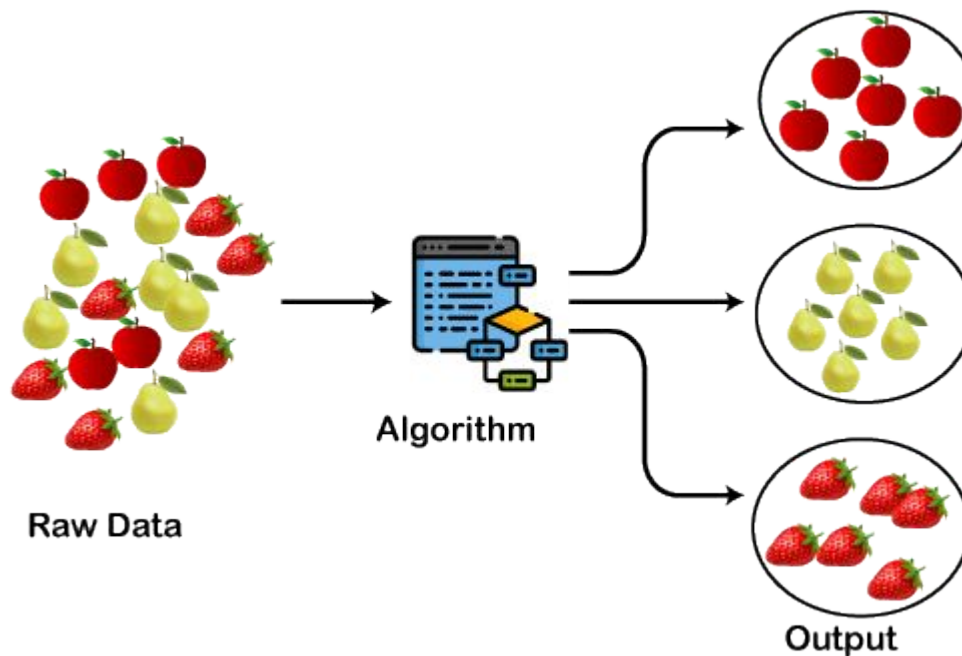
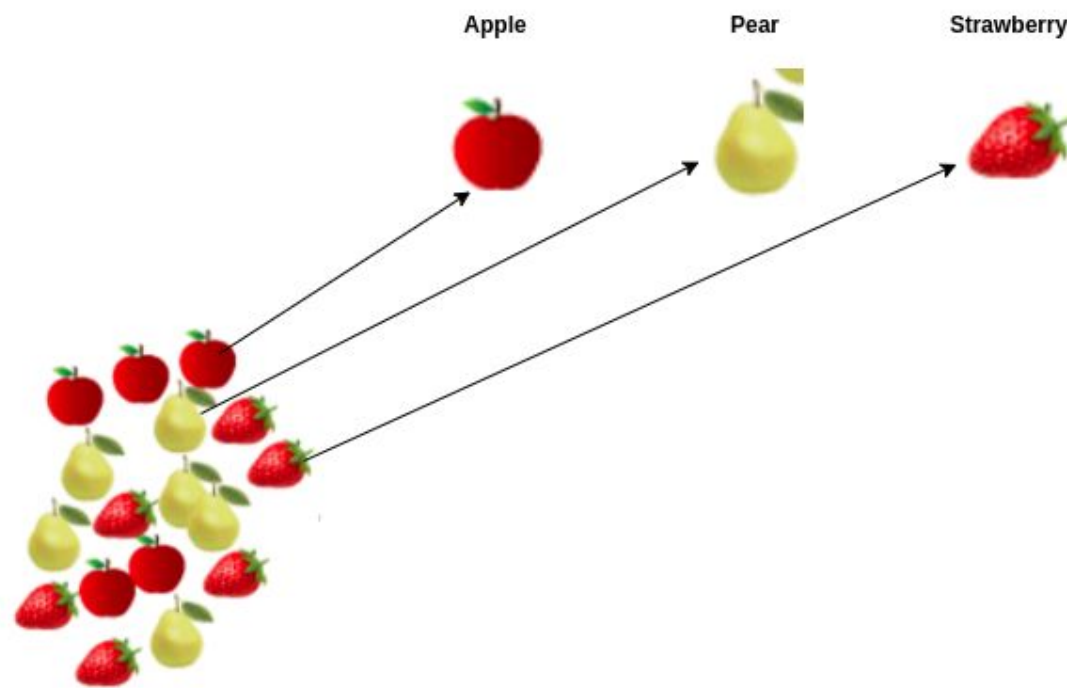


Image source: <https://www.javatpoint.com/clustering-in-machine-learning>

- Clustering: the process of organizing objects into groups of similar objects.
- A proximity (or distance) measure to allow one to assess if elements to be clustered are mutually similar or dissimilar.

Supervised learning



- Classification: We know in advance the categories.
- The model learns from the existing features to assign new object to a specific category.

Introduction

- Supervised learning: One of the fundamental paradigms in machine learning.
- An algorithm learns from labeled training data to make predictions or decisions without being explicitly programmed.
- Supervised learning forms the foundation of many real-world machine learning applications.
- A powerful approach for tasks where you have access to labeled data and want to automate decision-making or make predictions based on that data.

Introduction (2)

- The primary goal of supervised learning is to learn a mapping or relationship between the input data and the output labels.
- The model aims to make accurate predictions or classifications for new, unseen data based on what it has learned from the training dataset.
- Labeled data: A dataset consisting of input-output pairs. Each data point in the dataset includes both the input (features) and the corresponding desired output (label or target).
- Labels are given by humans through manual annotation.

Characteristics

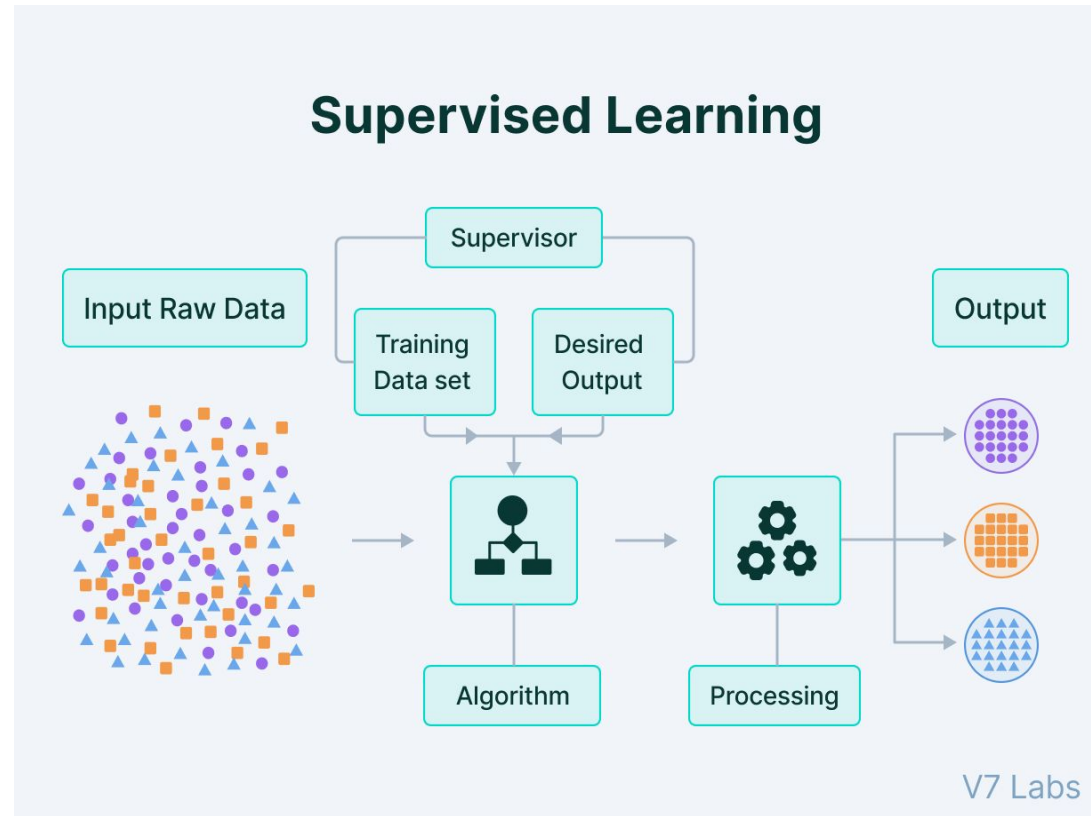


Image source: <https://www.v7labs.com/blog/supervised-vs-unsupervised-learning>

- Trained with labeled data.
- Main applications: Classification and regression.

Characteristics (2)

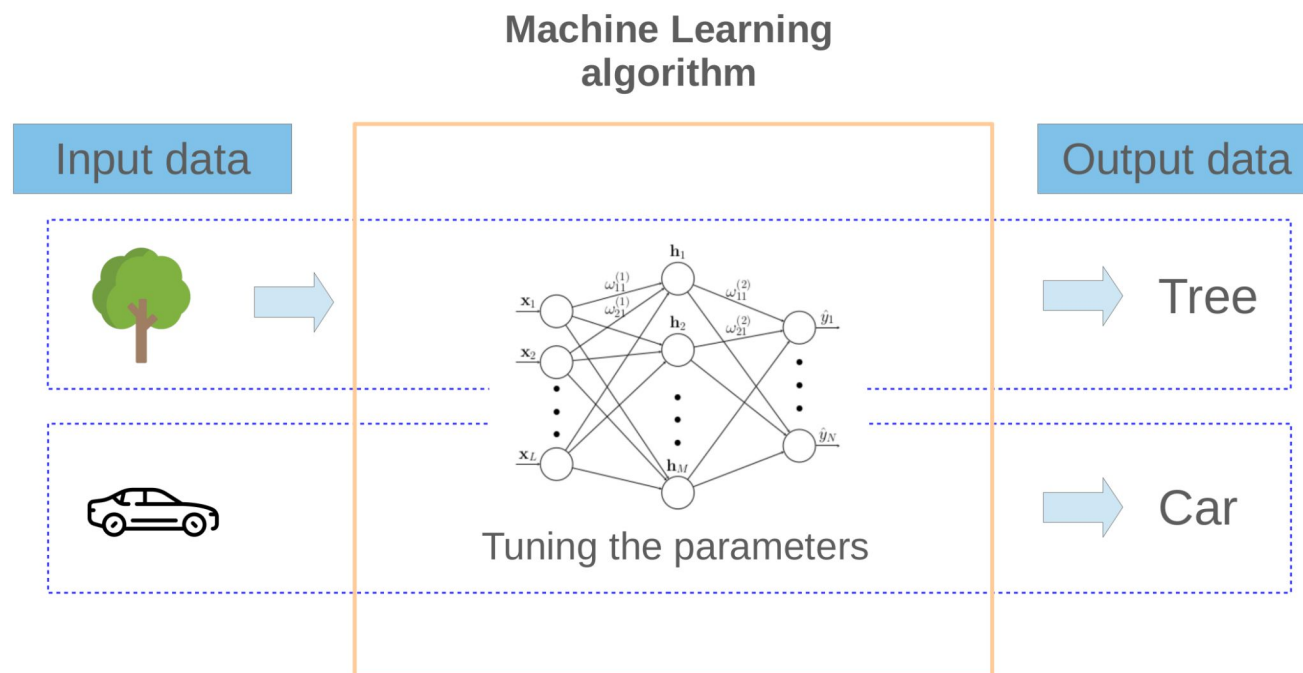
- Training phase: The algorithm uses the labeled training data to adjust its internal parameters and build a model that can make predictions.
- The learning algorithm seeks to minimize the difference between its predictions and the actual labels.
- Testing Phase: After training, the model is tested on a separate dataset (the test set) to evaluate its performance and assess its ability to generalize to new, unseen data.

How to train a Machine Learning classifier

— The learning process

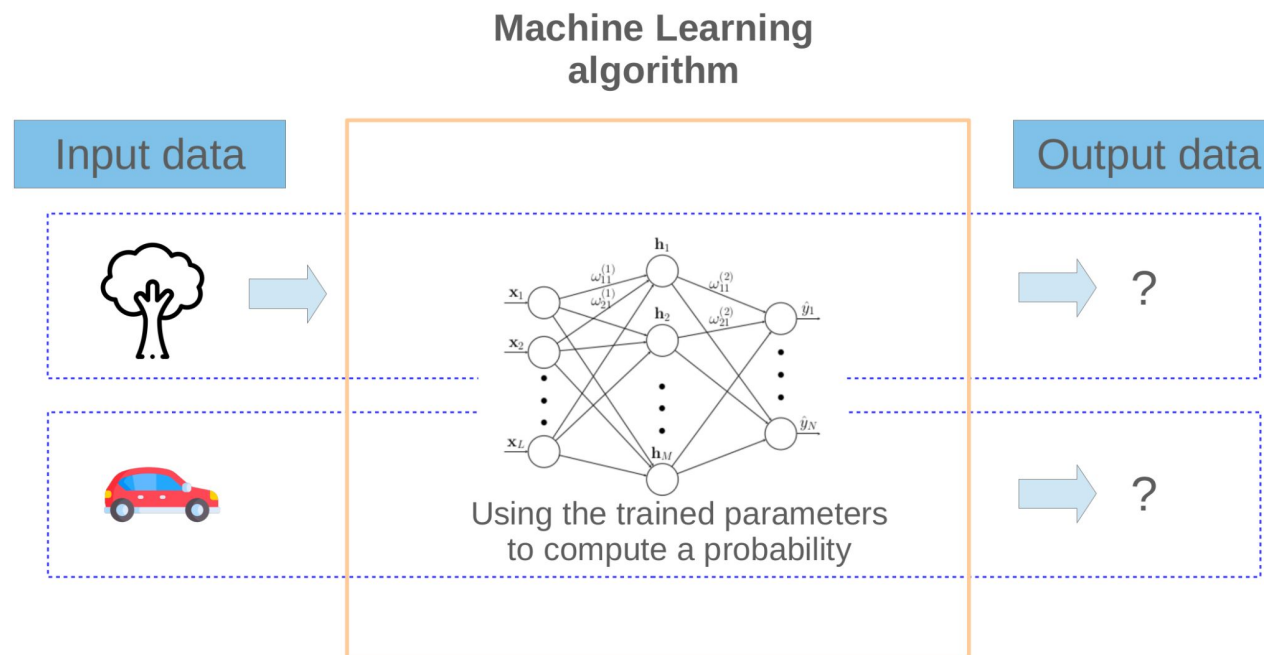
- Learning is actually the process of modifying the weights so that we can produce the desired outputs given the inputs.
- The more (good) input data you feed the network, the better accuracy you will obtain.

Training phase



- Feature: Input data (documents, images).
- Label: Given by humans.

Testing phase



- Feature: Input data (documents, images).
- Label: The algorithm tries to compute the probability, and give a label.

Train a dog

- Commands: "sit down", "shake hand".
- You press him, or grab his hand to instruct the corresponding action.
- Deployment: you speak out the command, and the dog does what he was taught.

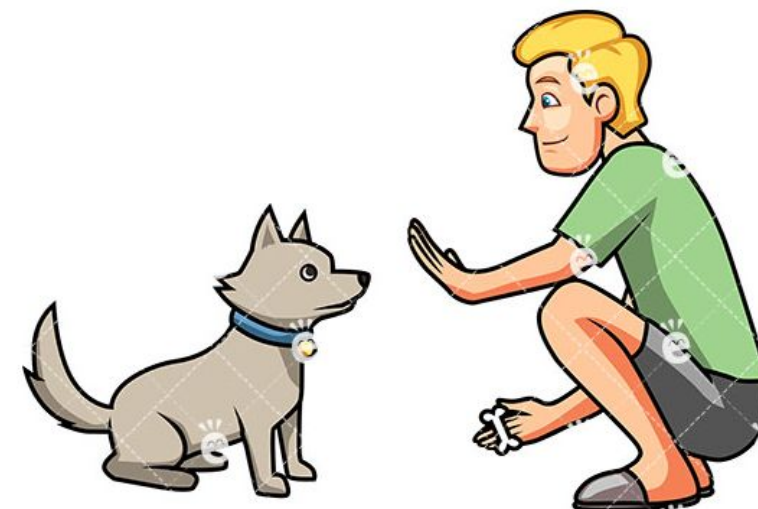
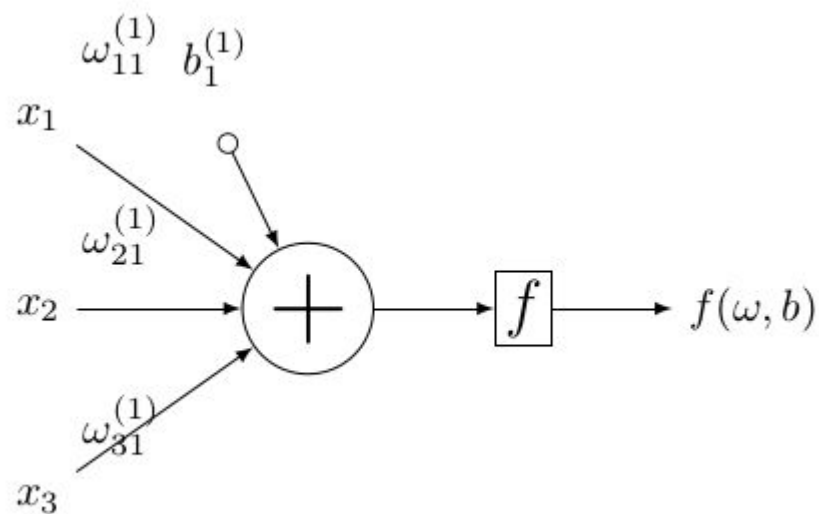


Image source: <https://friendlystock.com/product/man-training-his-dog-to-stay-giving-treat/>

Neural Network: Perceptron

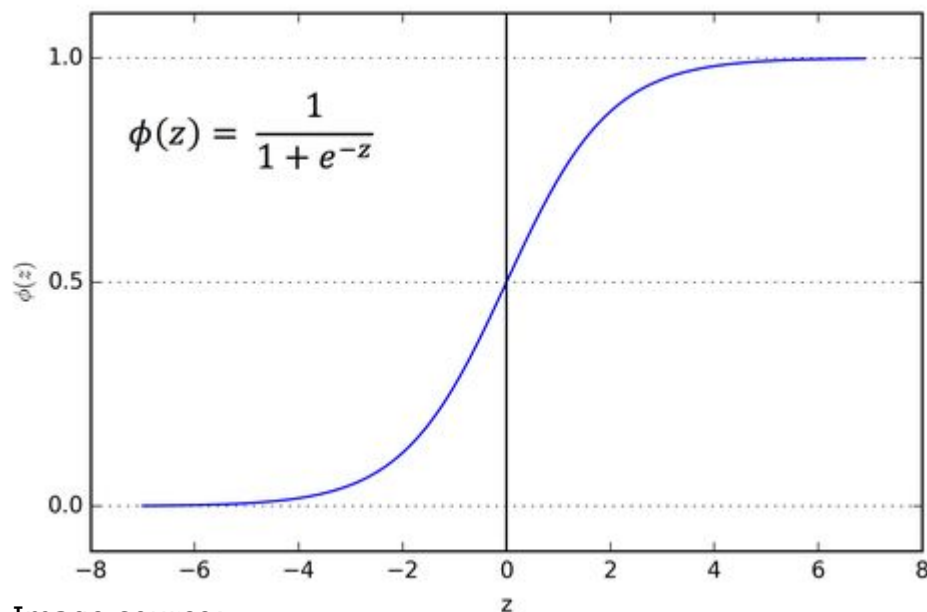


(a) A simple perceptron with 3 inputs

$$output = f\left(\sum_{i=1}^3 \omega_{i1}^{(1)} \cdot x_i + b^{(1)}\right)$$

- A decision making unit, taking into consideration different inputs

Activation: The sigmoid function

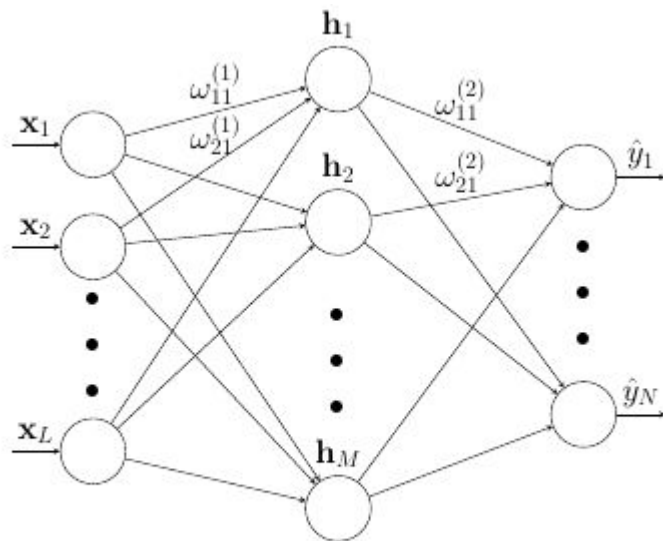


$$y = f(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + \exp(-\sum_j w_j x_j - b)}$$

Image source:
<https://towardsdatascience.com/activation-functions-neural-networks-1cbd9f8d91d6>

- Small changes in the weights (w) and biases (b) will trigger a small change in the output.

Feed-forward Neural Network



(b) A three-layer neural network

$$C(w, b) = \frac{1}{2n} \sum_x \|y(x) - (w.x + b)\|^2$$

$$\Delta y \approx \sum_j \frac{\partial y}{\partial w_j} \Delta w_j + \frac{\partial y}{\partial b} \Delta b$$

- Learning is actually the process of modifying the weights so that we can produce the desired outputs given the inputs.
- The more (good) input data you feed the network, the better accuracy you will obtain.

Learning

- Find a set of **weights and biases** (w, b) that minimize the cost function between the predicted values and the real ones using Stochastic Gradient Descent (SGD)



- Δy is a linear function of Δw and Δb

$$\Delta y \approx \sum_j \frac{\partial y}{\partial w_j} \Delta w_j + \frac{\partial y}{\partial b} \Delta b$$

Minimizing the cost function

- Find a set of weights that minimize the cost function between the predicted value and the real one using Stochastic Gradient Descent (SGD).

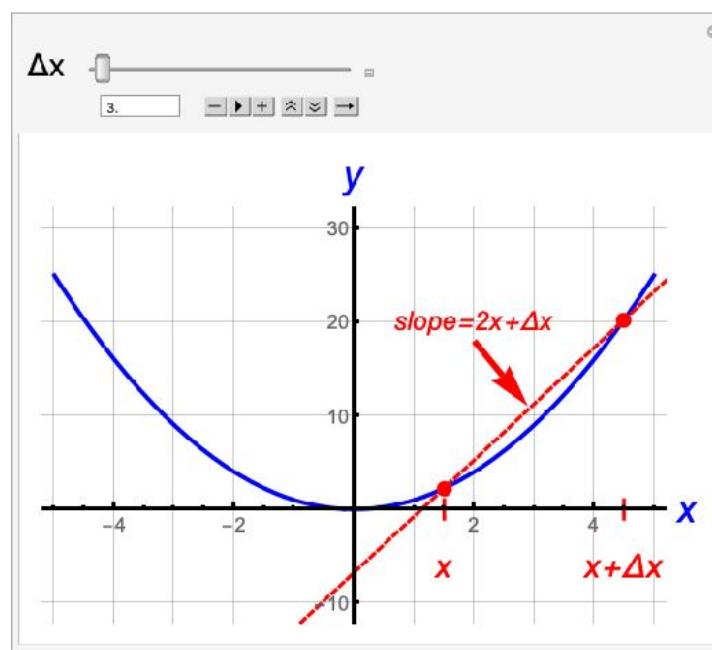


Image source: <https://infinityisreallybig.com/2019/09/20/the-derivative-of-y-x2/>

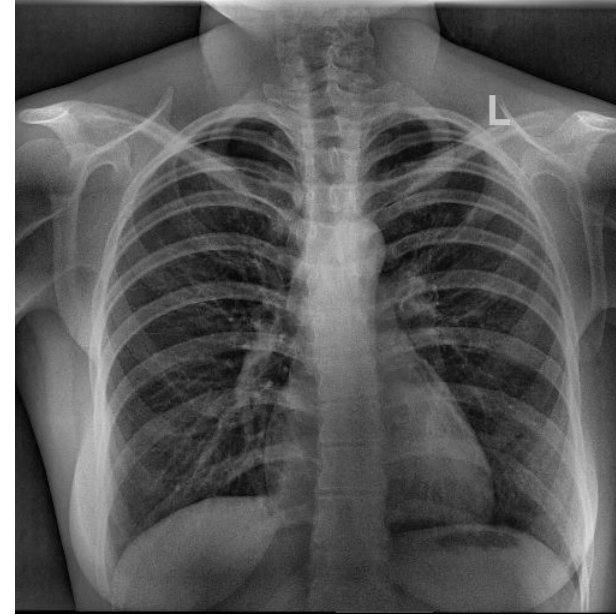
Overview

- Image and object recognition: Locating objects in a photo, detection of humans, objects.
- Natural language processing: Text classification, sentimental analysis (Twitter, Facebook, YouTube).
- Machine translation: Translating a document in an original language into a document in a different language (Google translate).
- Writing assistance: Google Docs, Grammarly.

Overview (2)

- Healthcare: Disease diagnosis, detection of Covid ([our paper](#)), detection of tuberculosis ([our paper](#)), detection of electrocardiogram signals ([our paper](#)).
- Finance: Deep learning models are used for fraud detection.
- Autonomous (driveless) vehicles: Object detection, lane tracking, decision-making (Tesla).
- Google Maps: Segmentation of car plates, human faces.

Diagnostic Imaging



- Labelled images are fed as input for deep neural networks.
- The trained systems are used for the recognition of unknown images.

Diagnostic Imaging (2)

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Automatic detection of Covid-19 from chest X-ray and lung computed tomography images using deep neural networks and transfer learning

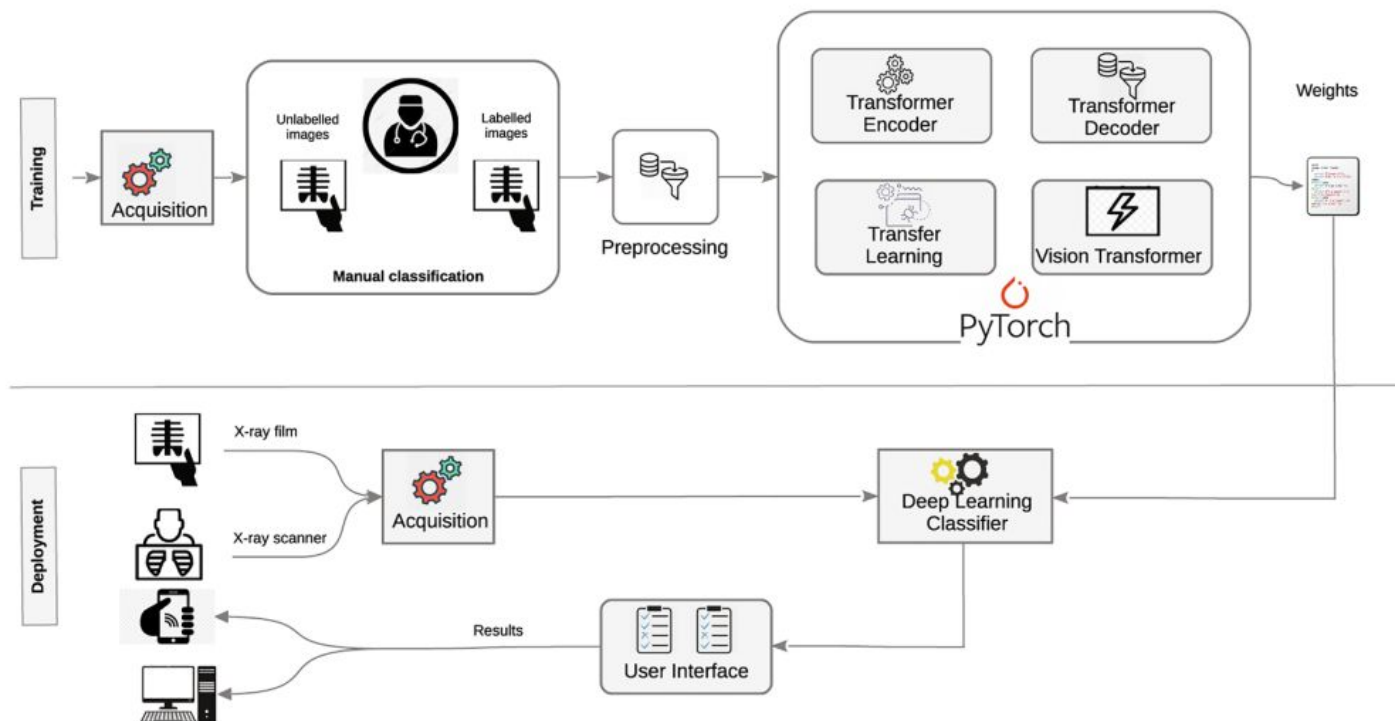
Linh T. Duong^a, Phuong T. Nguyen^{b,*}, Ludovico Iovino^c, Michele Flammini^c

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Diagnostic Imaging (3)



- A system built on top of Vision Transformer.
- The trained systems are used for the recognition of unknown images.

Diagnostic Imaging (4)

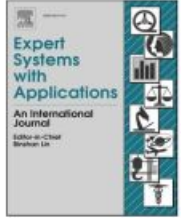
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Detection of tuberculosis from chest X-ray images: Boosting the performance with vision transformer and transfer learning

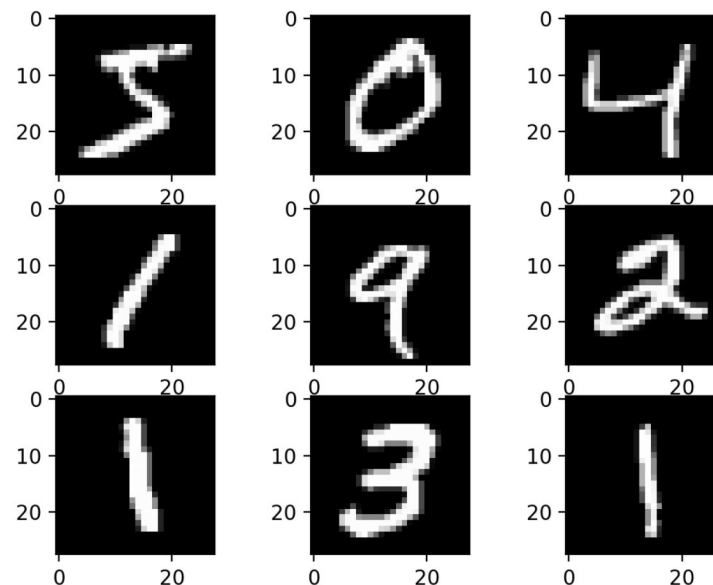
Linh T. Duong^a, Nhi H. Le^a, Toan B. Tran^a, Vuong M. Ngo^b, Phuong T. Nguyen^{c,*}

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DOI: <https://doi.org/10.1016/j.eswa.2021.115519>

Recognition of handwritten numbers



Plot of a Subset of Images From the MNIST Dataset

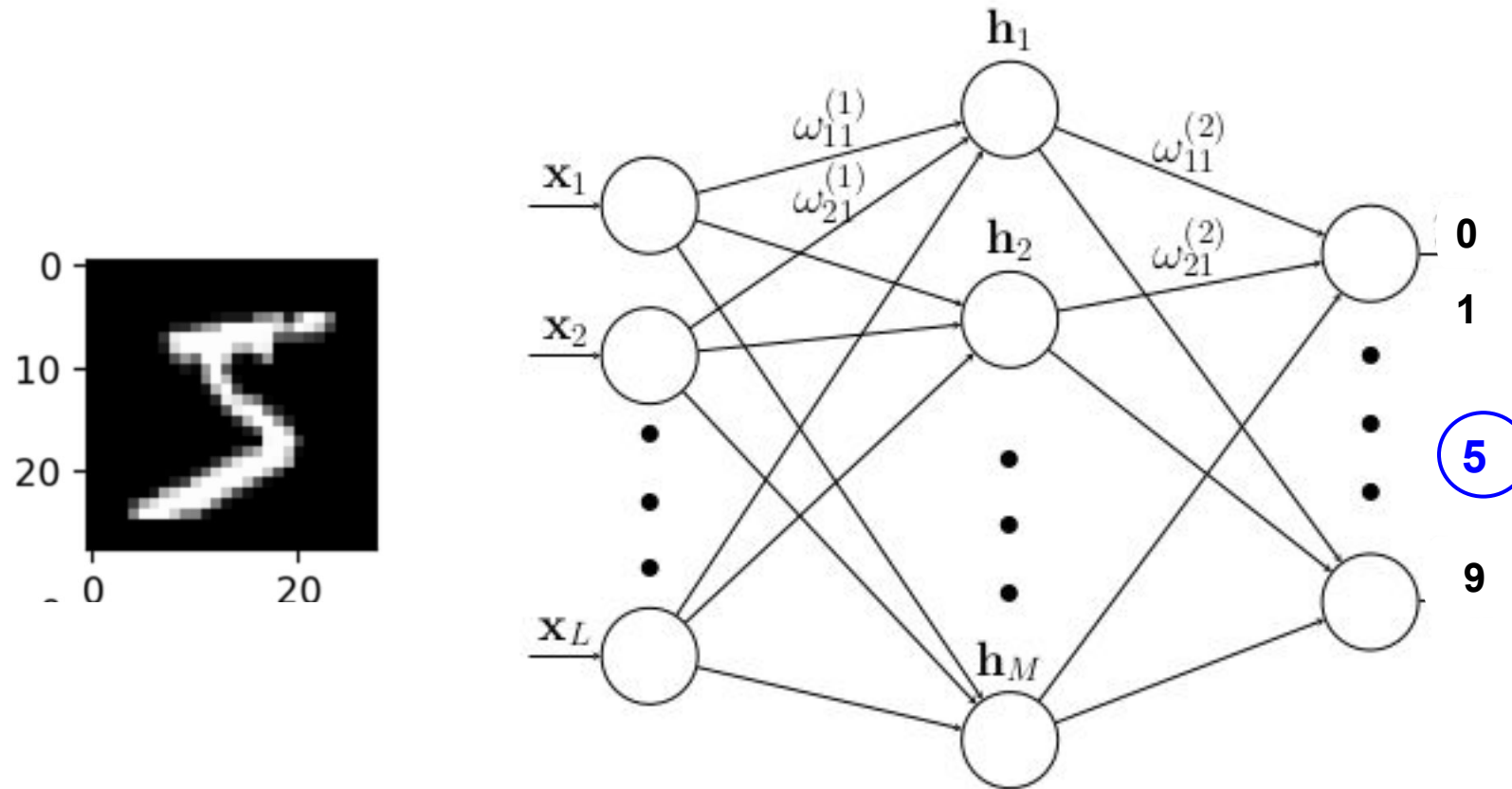
Image source: bit.ly/3tfjKah

- MNIST: A dataset with 60,000 training and 10,000 testing images.
- Humans were involved in manually writing the numbers, and giving the corresponding label.

Recognition of handwritten numbers

- Training: a set of images with 10 labels: from 0 to 9.
- Deployment: an image containing a handwritten number is fed into the network, which computes and returns the predicted number.
- Each image is a 28 x 28 pixel matrix.
- Each image is represented in a row, the first column is the label, and the remaining 784 columns contain grayscale level of one pixel (from 0-255).

Application: Classification of handwritten numbers



Driverless cars

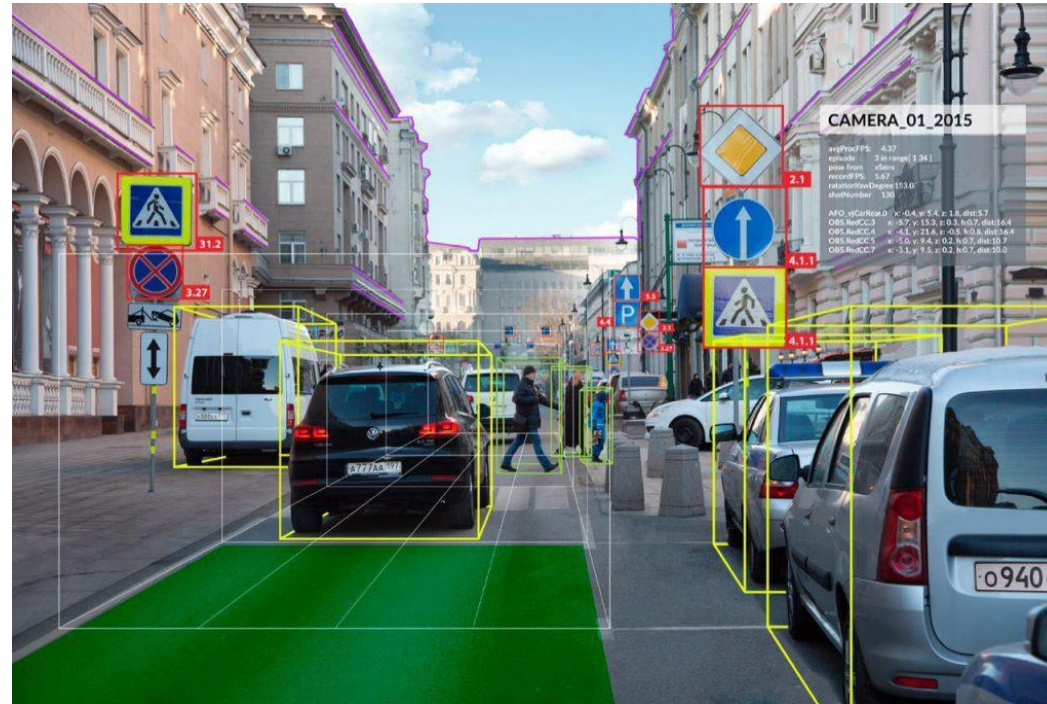


Image source:

<https://becominghuman.ai/computer-vision-applications-in-self-driving-cars-610561e14118>

- Deep neural networks detect traffic lights, road signs, people.
- Decisions are made to swiftly react to the surrounding environment.

Driverless cars (2)

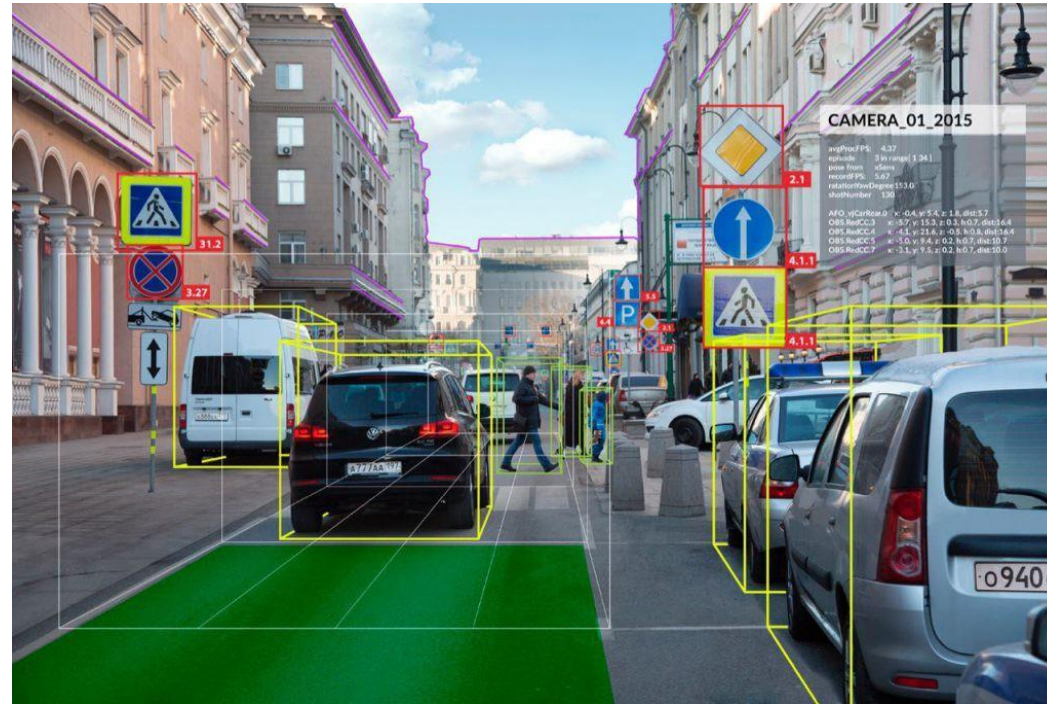


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- Segmentation of objects: Deep Learning models trained on large image datasets.
- Detection of individual objects from captured images.
- Input features: Image; Output feature: Labels (Car, person, tree, building, etc.)

License plate recognition

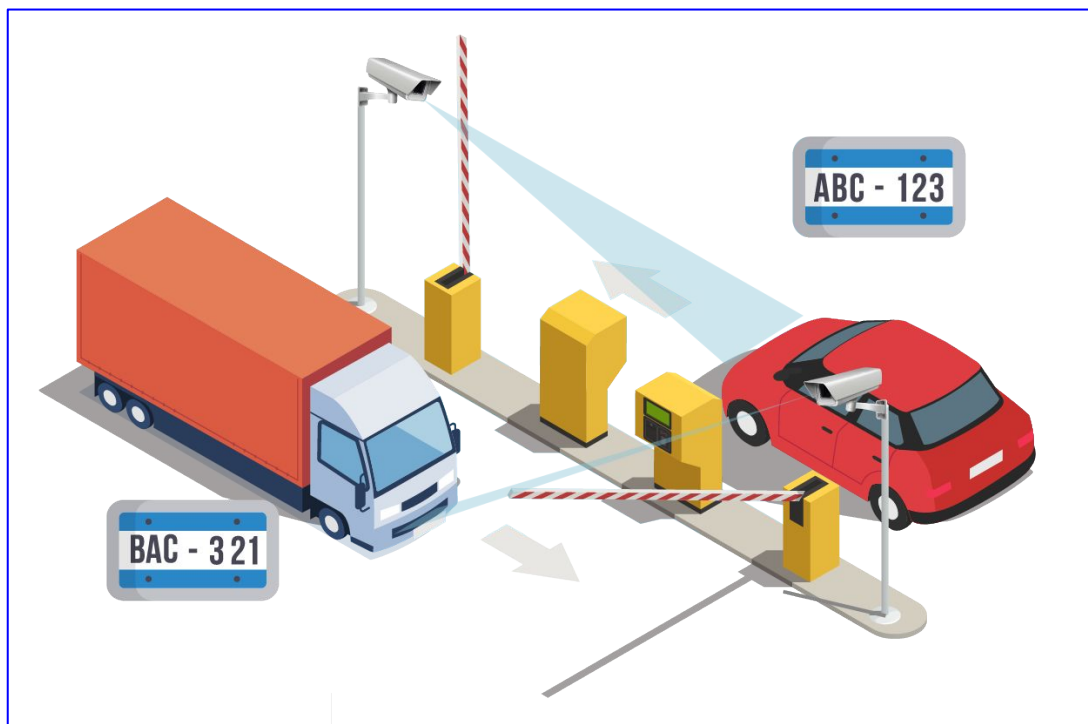
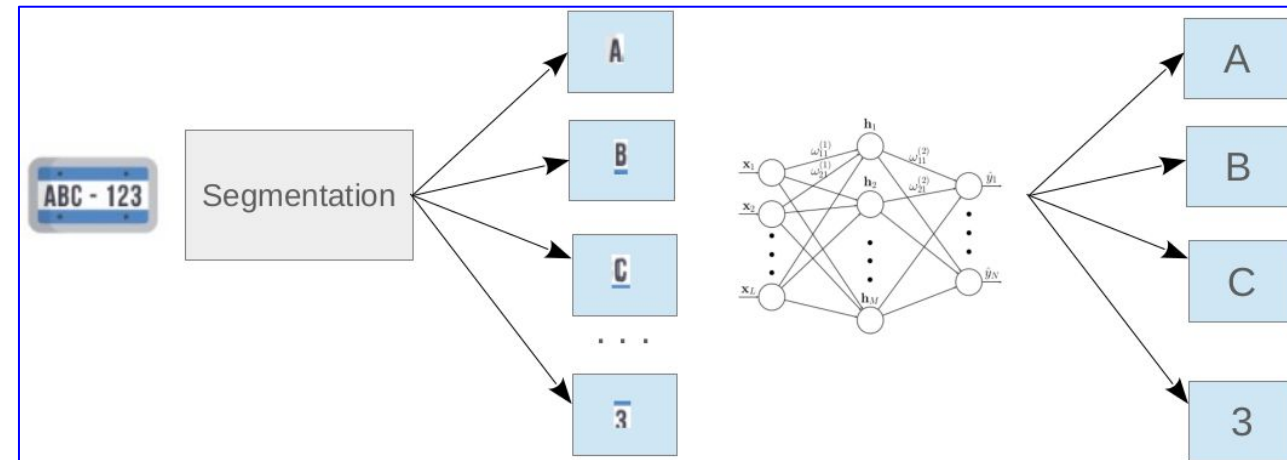


Image source: <https://bit.ly/3ZLNzvd>



- Capture license plate from moving vehicles.
- Segmentation and detection of numbers, characters.
- Features: input images, segmented to get individual characters, numbers.

License plate recognition



- Google uses supervised learning to blur humans faces, and cars license plates.